Ku-Yu Fan

Ph.D. Candidate in Mathematics | Nagoya University

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RESEARCH INTERESTS

Multiple zeta values; motivic periods and coactions; Schur multiple zeta values; arborified multiple zeta values; p-adic multiple zeta values; multiple polylogarithms; p-adic multiple L-functions.

EDUCATION

2024-present **Ph.D. in Mathematics | Nagoya University**
Graduate School of Mathematics - Expected September 2027
Advisor: Hidekazu Furusho

2021-2024 **M.Sc. in Mathematics | National Taiwan University**
Institute of Mathematics
Advisor: Nobuo Sato

2017-2021 **B.Sc. in Mathematics | National Tsing Hua University**
Department of Mathematics

REFEREED RESEARCH ARTICLES

2026 **Coaction formula for the motivic version of Yamamoto's integral**
Mathematische Zeitschrift 314, Article 35 (2026). DOI: 10.1007/s00209-026-04138-w

2026 **A map between arborifications of multiple zeta values**
Journal of Algebra (2026). DOI: 10.1016/j.jalgebra.2026.06.006

PREPRINTS

2026 **Regularised double shuffle relations for planar Arborified Zeta Values**
Pierre Catoire, Pierre J. Clavier and Ku-Yu Fan. arXiv:2609.02346

2026 **p-adic multiple L-functions and twisted multiple Bernoulli numbers**
Ku-Yu Fan. arXiv:2607.04086

2026 **p-adic multiple zeta values of integer indices**
Ku-Yu Fan. arXiv:2603.22923

EXPOSITORY PUBLICATION

2021 **The Number of Lattice Lines in Lattice Figures [in Chinese]**
Ku-Yu Fan and Feng-Mei Lin. Mathmedia (Shuxue Chuanbo) 45(2), 44-53 (2021).

SELECTED RESEARCH TALKS

Sep 2026	**Explicit Formulas for Cyclotomic Multiple Bernoulli Numbers**	NCTS Number Theory Seminar, National Tsing Hua University
Sep 2026	**An explicit formula for the special values of the p-adic multiple L-functions**	Mathematical Society of Japan, Autumn Meeting, Kobe University
Jul 2026	**Double shuffle relations for p-adic multiple zeta values of integer indices**	5th International Workshop on Multiple Zeta Values and Related Topics, Central South University
Jul 2026	**Double shuffle relations for integer indices and their application to p-adic multiple zeta values**	25th Sendai-Hiroshima Number Theory Conference, Tohoku University
Jun 2026	**p-adic multiple zeta values and double shuffle relations for integer indices**	Algebraic and Combinatorial Perspectives in the Mathematical Sciences, online
Mar 2026	**p-adic multiple zeta values of integer indices**	Meeting in the Middle (Characteristic 0 and p Multiple Zeta Values), University of Hawaii at Manoa
Feb 2026	**p-adic multiple L-function and twisted multiple Bernoulli numbers**	MATHSCI Freshman Seminar, Nagoya University
Sep 2025	**A Map Between Arborifications of Multiple Zeta Values**	RIMS Symposium on Multiple Zeta Values, Kyoto University
Sep 2025	**A Map Between Arborifications of Multiple Zeta Values**	Mathematical Society of Japan, Autumn Meeting, Nagoya University
Jul 2025	**A Map Between Arborifications of Multiple Zeta Values**	4th International Workshop on Multiple Zeta Values, Tongji University
Mar 2025	**A map between arborifications of multiple zeta values**	Monthly Meeting, Nagoya University (blackboard talk)
Nov 2024	**Coaction Formula for the Motivic Version of Yamamoto's Integral**	2nd Kindai Workshop: Multiple Zeta Values and Modular Forms, Kindai University
May 2024	**Coaction formula for the motivic version of Yamamoto's integral**	Monthly Meeting, online

TEACHING EXPERIENCE

Teaching assistantships; recurring courses grouped by institution, all semesters preserved.

TA — **Nagoya University**
Special Topics in Foundations (Analysis) - Spring 2026;
Calculus - Spring and Fall 2025

TA — **National Taiwan University**
Calculus - Fall 2022, Spring and Fall 2023, Spring 2024;
Modern Algebra - Fall 2022, Spring 2023;
Linear Algebra - Fall 2021, Spring 2022

TA — **National Tsing Hua University**
Elementary Number Theory - Spring 2021;
Calculus - Fall 2020, Spring 2021;
Advanced Calculus - Spring 2020

FELLOWSHIPS AND AWARDS

2024-2027 — **Japan-Taiwan Exchange Association Scholarship**
Graduate Research Students Fellowship (October 2024-September 2027).

2019 — **Professor Han-Ching Lai Memorial Scholarship**
Department of Mathematics, National Tsing Hua University.

ACADEMIC SERVICE & ORGANIZATION

2026 — **Organizing Committee - Nagoya-Kansai Young Researchers' Zeta Meeting**
Joint meeting of the 19th Young Mathematicians Conference on Zeta Functions and 70th Kansai Multiple Zeta Seminar; supported by a selected Nagoya University student project (2025).

Sep 2024- — **Organizer and Representative - Monthly Meeting**
A monthly research meeting on multiple zeta values.

SELECTED PATENTS

2021 — **Single stacking element and component structures for multi-style combinations**
Taiwan utility model M612210.

2018 — **Rhombic dodecahedron puzzle and multiple rhombic dodecahedron puzzle**
Taiwan utility model M566100; US Patent 10,722,782.

2017 — **Hexagonal prismatic packing puzzle**
Taiwan utility model M548578; US Patent 10,322,337.

All listed patents co-invented with Wen-Liang Hung.